

Accuracy comparison for prosthesis activation using a commercial sEMG sensor and an FMG sensor based on FBG and 3D-printing

Felipe R. Cortes^a, María Gaitán-Padilla^a, Marcelo E. V. Segatto^a, Maria José Pontes^a, Carlos A. Cifuentes^b, and Camilo A. R. Díaz^a

^aTelecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo (UFES), Vitória, Brazil

^bBristol Robotics Laboratory, University of the West of England, Bristol, United Kingdom.

ABSTRACT

Myoelectric information has been recently used for actuated prostheses control. The most used approaches are surface electromyography (sEMG) and force myography (FMG). Recent works have exhibited FMG sensors with high pattern recognition accuracy compared with sEMG, although with a better signal-to-noise ratio, and less impact of sweating. In this context, optical fiber sensors have been used as FMG by measuring the volumetric change of the muscle during force contraction, which can be achieved because of their high sensitivity to strain, flexibility, and robustness. This work aims to generate a preliminary force and angle classification algorithm for prosthesis control by comparing different machine-learning models using sEMG and FMG. For this purpose, six volunteers participated in force exertion tasks while the grip force and the hand closure angle were measured. Results showed that the RF model obtained the best accuracy (96 and 98 %) and precision (97 and 98 %) in the angle and force classification, respectively. This approach showed the potential use of optical fiber FMG sensors, combined with sEMG, for prostheses control by implementing machine learning models. Further studies will consider more volunteers and features to make the algorithm more robust and improve the results to control a prosthetic device.

Keywords: sEMG, FMG, FBG, Optical Fiber Sensor, Machine Learning.

1. INTRODUCTION

Amputation is used as a treatment when a limb suffers an irreversible loss of circulation or functionality.¹ This procedure significantly impacts a person's independence and quality of life; in the case of the upper limb, it affects the grasping function, which is essential for activities of daily living.² Bei Yuan *et al.* revealed that traumatic amputations increased to 13.23 million in 2019 considering 204 countries, highlighting the importance of this concern.^{3,4} In addition, amputees have social and health care needs, including rehabilitation, adaptive self-care training, and the integration of prosthetic devices to complement limb function.⁵

In addition, prostheses are divided into active and passive devices, with active prostheses including various intelligent and manually operated systems,⁶ while passive prostheses are mainly for aesthetic purposes. Usually, actuation is based on sensors and machine learning algorithms, where real-time information about the intention of movement (i.e. angle, muscle contraction, grasping type, and force) is used to control the device, e.g. robotic prostheses actuated by myoelectric data⁷ and body-based information.⁸

Further author information: (Send correspondence to F.R.C.)

F.R.C.: felipe.cortes@edu.ufes.br

M.G.P.: maria.padilla@edu.ufes.br

M.E.V.S.: marcelo.segatto@ufes.br

M.J.P.: maria.pontes@ufes.br

C.A.C.: Carlos.Cifuentes@uwe.ac.uk

C.A.R.D.: camilo.diaz@ufes.br

The most common myoelectric actuation systems use surface electromyography (sEMG) and force myography (FMG). More recently fiber optic sensors have been adopted for robotic control because of their immunity to electromagnetic noise, sensitivity, flexibility, potential use in embedded systems, and affordability.^{9,10} In addition, the control loop depends on robust algorithms that use the sensor information to operate the prostheses, resulting in correct performance and positive attitudes towards using robotic and sensor-controlled devices.¹¹

While sEMG sensors measure the combined effect of neuromuscular electrical activity on the skin surface, FMG sensors measure pressure differences due to volumetric changes in muscle.¹² However, sEMG requires electrodes that can deteriorate with daily use, sweat, and humidity, and requires careful positioning on the target muscle surface to activate the assistive device properly. In addition, sEMG can lead to fatigue with prolonged use, which can have a negative effect on the intention to use and perception of active prostheses.¹³ Nevertheless, while FMG sensing solves most of those issues, it can also be influenced by muscle strength and size, which may be reduced as a result of amputation.

Nowadays, approaches that mix both signals have been implemented to compensate for fatigue issues, muscle size, and daily tasks.^{14–16} In this sense, this work aims to generate a preliminary force and angle classification algorithm for prosthesis control by comparing different machine-learning models with performance metrics. This was achieved using a commercial sEMG sensor and an optical fiber-based FMG sensor on six volunteers performing a muscle contraction task, measuring the angle of the fingers and the force exerted.

2. MATERIALS AND METHODS

This work was approved by the ethics committee of the Federal University of Espirito Santo (report number: 6847995). Six volunteers participated (two women and four men, 23 ± 3 years old). Firstly, the protocol was explained and all the questions were answered by the researchers, after that, the participants signed the informed consent. Then, the sEMG and FMG sensors were placed to monitor the flexor digitorum superficialis (FDS) and extensor digitorum superficialis (EDS) muscles of the participants as shown in Figure 1 (a). In addition, three red markers were placed on the metacarpophalangeal (MCP) joint, the proximal interphalangeal (PIP) joint, and the fingertip of the users, to calculate the angle, as represented in green in Figure 2b. Participants were asked to sit and be comfortable to conduct the force exertion task, while two cameras were used to record the force of the dynamometer and to measure the angle with the markers, as presented in Figure 1 (b). Then, they were asked to perform three levels of force divided into high, medium, and low, then the dynamometer grip was adjusted to decrease the finger angle three times per participant.

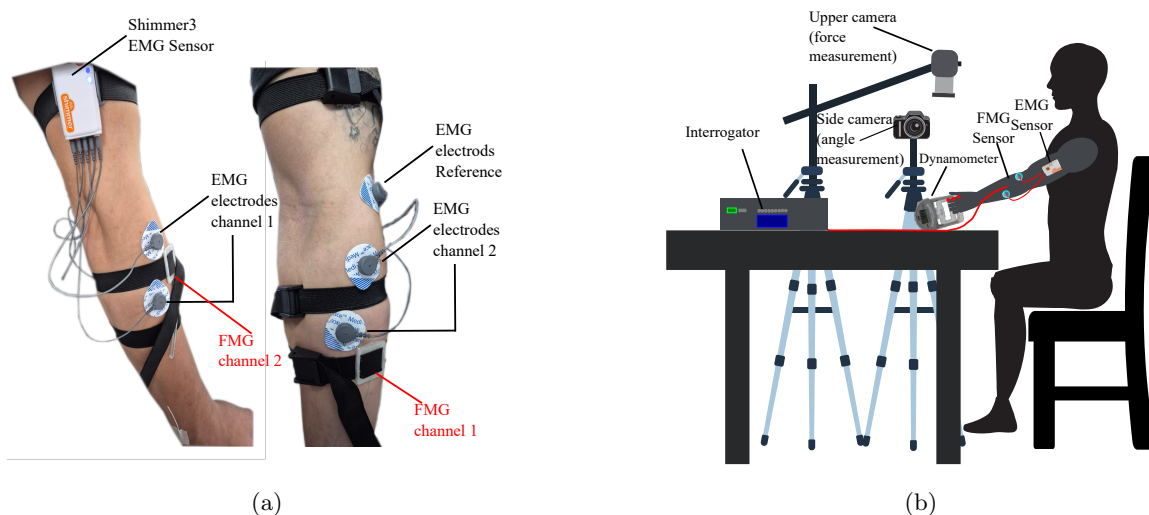


Figure 1: Detailed protocol. (a) Sensors placement on the forearm and (b) Experimental setup.

For the sEMG lecture, a commercial EMG sensor (Shimmer3 ECG/EMG, Ireland) was used, this device allows the lecture of two EMG channels using an Analog-to-digital converter (ADC) of 24-bits and the wireless

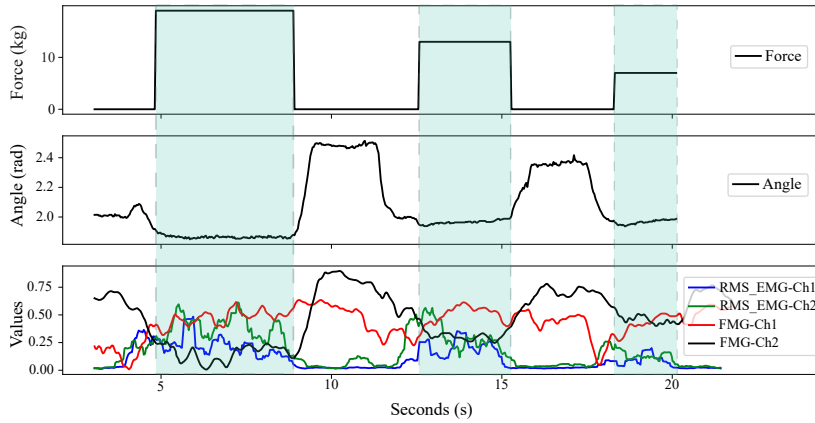
data sending at a baud rate of 1024 Hz using Bluetooth 4.2. In addition, for the conduction of the study it was used foam electrodes with adhesive hydrogel (Kendall, covidien), a dynamometer (DM-90, INSTRUTHERM) for the force handling measurement, and the Kinovea software for the angle estimation.

The FMG sensors were developed following the steps explained by Cortes *et al.*,¹⁷ with a single FBG with a centered Bragg wavelength of 1547 nm and 3 mm long. The FBGs were inscribed using the phase mask method in a photosensitive single-mode optical fiber (GF1, Thorlabs) using a laser with a frequency of 10 Hz, energy of 0.37 mJ, and centered at 265.524 nm (DPS-266-Q, Changchun New Industries). The data were collected using a four-channel interrogator (HYPERION si255, Luna's) with a wavelength range from 1500 to 1580 nm and a sampling rate of 5000 Hz.

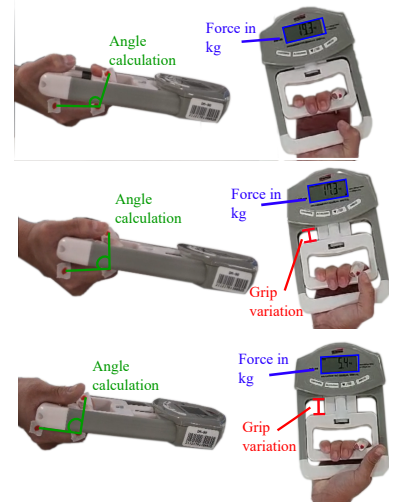
The sensor data were down-sampled to 30 Hz for further synchronization with the videos, the angle was calculated using the point data extracted from the Kinovea software with the Equation 1, where α is the angle in radians, d_1 the distance between the MCP and the PIP, d_2 the distance between the PIP and the fingertip, and d_3 the distance between the MCP and the phalanx.

$$\alpha = \text{ARCCOS} \left(\frac{d_1^2 + d_2^2 - d_3^2}{2d_1d_2} \right), \quad (1)$$

Once the angles were calculated and the data synchronized, the EMG and FMG values were normalized considering the maximum voluntary contraction values (MVC). In the case of the FMG signal the minimum value is at rest, and for the EMG signal the root mean square (RMS). The dataset was created by obtaining the mean in 10-sample windows of the RMS values of the EMG signal, the FMG signal, and the angular values during contractions approximated as a natural number, adding also the maximum force in kilograms measured for the dynamometer during that contraction approximated as a natural number. In the end, the dataset contained 848 rows to train and validate different models of machine learning algorithms.



(a)



(b)

Figure 2: Sensor Monitoring and Grasp Adjustment Analysis. **(a)**. Force, angle, and sensors correlation example, where the FMG1 and EMG2 monitored the EDS muscles, and FMG2 and EMG1 monitored the FDS muscles. **(b)**. Angle comparison for different grasp adjustments in the dynamometer.

The machine learning classifiers were K-Nearest Neighbors (KNN), Gaussian Naive Bayes (GNB), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM), and the K-Fold cross-validation was used with $k = 5$ to obtain the accuracy and precision of each model.

3. RESULTS

An example of the signals is shown in Figure 2a. FMG1 and EMG2 monitored the EDS muscles and FMG2 and EMG1 monitored the FDS muscles, it can be seen the activation of the four channels in the force exertion task and the three force levels. In the case of the angle, it was inverse to the force applied by closing the hand during the grasp, and it can be seen a difference between the high and medium force levels, however, smaller differences between the medium and low force exertion were presented. Finally, the time for the execution of the force was free for the subjects and there is a tendency to reduce the time for the execution of the force in the three levels. The resistance of the dynamometer was adjusted three times per participant, generating a bigger variation range for the angle, as shown in Figure 2b.

After training the six models for angle and force classification, the mean and standard deviation of the accuracy and precision in cross-validation were obtained, as shown in Table 2, and Table 1. In addition, the hyperparameters were adjusted to obtain the best response of the algorithm. The RF was the best-performing algorithm for angle with the hyperparameters $n_estimators = 200$, $max_depth = 20$, $min_samples_split = 4$ and $min_samples_leaf = 2$, and force estimation with $n_estimators = 300$, $max_depth = 20$, $min_samples_split = 4$ and $min_samples_leaf = 2$, comparing accuracy and precision and using the four channels. These results showed the complementarity between EMG and FMG signals in both classification tasks. Figure 3a shows the confusion matrix of a one-way result for angle classification with the RF algorithm, and Figure 3b displays the confusion matrix of the RF model for force estimation.

Table 1: Average accuracy and precision across folds for angle classification.

Model	Angle classification (value $\pm$ standard deviation %)					
	EMG-RMS		FMG		Both	
	Acc	Pre	Acc	Pre	Acc	Pre
KNN	46 $\pm$ 4.0	46 $\pm$ 4.7	59 $\pm$ 4.0	59 $\pm$ 4.3	78 $\pm$ 2.6	80 $\pm$ 2.1
GNB	25 $\pm$ 4.9	18 $\pm$ 6.6	35 $\pm$ 3.5	29 $\pm$ 3.6	40 $\pm$ 3.4	41 $\pm$ 3.0
LR	24 $\pm$ 3.3	13 $\pm$ 7.8	32 $\pm$ 3.5	25 $\pm$ 3.6	39 $\pm$ 5.0	33 $\pm$ 5.2
DT	50 $\pm$ 4.5	48 $\pm$ 5.5	64 $\pm$ 3.5	66 $\pm$ 2.9	74 $\pm$ 2.3	74 $\pm$ 2.4
RF	56 $\pm$ 4.0	55 $\pm$ 4.7	63 $\pm$ 4.4	66 $\pm$ 4.7	96 $\pm$ 0.6	97 $\pm$ 0.4
SVM	42 $\pm$ 5.6	42 $\pm$ 5.9	53 $\pm$ 3.0	55 $\pm$ 2.7	75 $\pm$ 3.2	77 $\pm$ 3.5

Table 2: Average accuracy and precision across folds for force classification.

Model	Force classification (value $\pm$ standard deviation %)					
	EMG-RMS		FMG		Both	
	Acc	Pre	Acc	Pre	Acc	Pre
KNN	42 $\pm$ 2.1	43 $\pm$ 2.1	51 $\pm$ 1.6	51 $\pm$ 3.4	71 $\pm$ 2.7	73 $\pm$ 3.0
GNB	23 $\pm$ 3.2	15 $\pm$ 1.8	28 $\pm$ 2.4	23 $\pm$ 3.0	42 $\pm$ 3.8	39 $\pm$ 5.2
LR	19 $\pm$ 3.1	10 $\pm$ 2.5	18 $\pm$ 3.9	10 $\pm$ 3.6	31 $\pm$ 3.3	24 $\pm$ 4.2
DT	29 $\pm$ 3.6	28 $\pm$ 3.8	68 $\pm$ 2.6	71 $\pm$ 3.3	85 $\pm$ 1.5	87 $\pm$ 1.8
RF	52 $\pm$ 3.2	54 $\pm$ 4.4	75 $\pm$ 2.1	77 $\pm$ 2.9	98 $\pm$ 0.9	98 $\pm$ 0.9
SVM	33 $\pm$ 2.4	29 $\pm$ 3.6	50 $\pm$ 2.9	53 $\pm$ 4.9	78 $\pm$ 2.7	79 $\pm$ 2.2

4. DISCUSSION

This paper shows the comparison of the accuracy of six commonly used machine learning classifiers¹⁸ using the RMS data of the EMG signal, the FMG signal, and both signals for training and validation to compare with performance metrics. The RF algorithm obtained the highest overall accuracy values of all classifiers when the two signals were combined, and it can be observed that the accuracy using only the EMG RMS or the FMG was significantly lower than when both were used. In addition, the accuracies obtained for force detection compare well with the approach of Jiang *et al.*,¹⁹ who obtained the best accuracy with a support vector regression and a linear discriminant analysis classifier of 100% in validation and 96% in testing, performing complex grasping

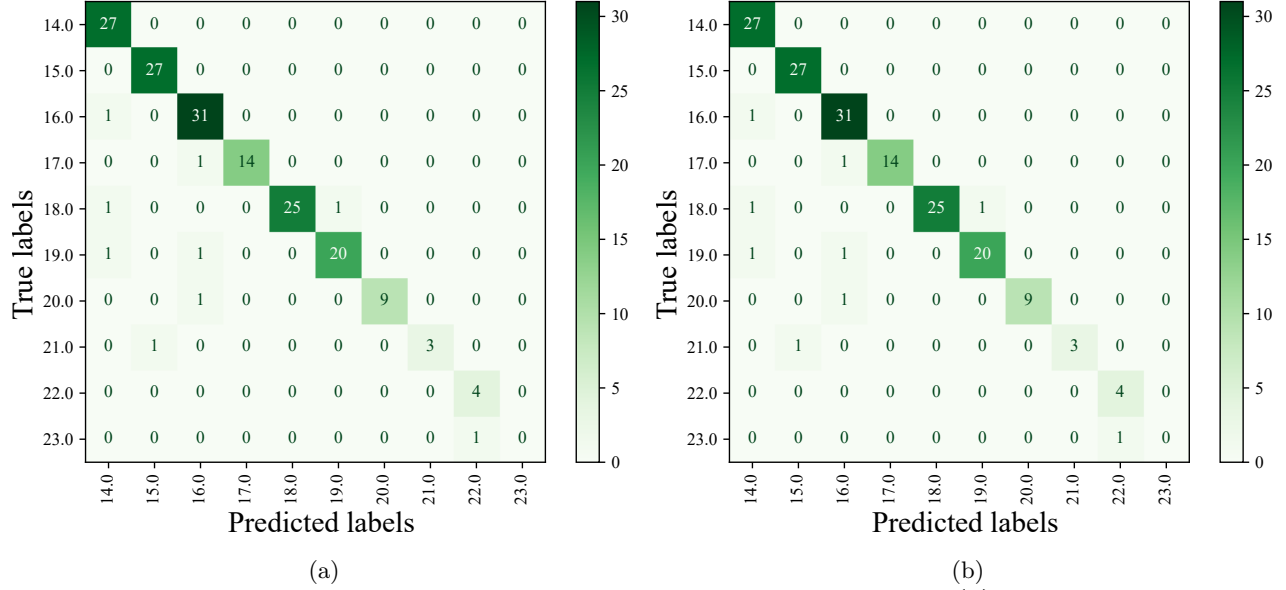


Figure 3: Confusion matrix of one fold using the best-performed algorithm RF. **(a)**. For angle classification. **(b)**. For force classification.

tasks and using a dynamometer for the force reference values in eight force levels. The prediction of grasp types can be a further work combining both sensors and performing the activation of the available prosthetic device,²⁰ adjusting the prosthetic fingers angles with the predicted angle and force.

Due to the approximations, the classifiers trained for force classification achieved a resolution of 1 kg, whereas the classifiers trained for angle classification achieved a resolution of 0.1 radians. In addition, some other variables must be taken into account as perceived in this protocol, such as the muscle fatigue presented during the force exertion and after the three trials of the three force tasks asked, which can be identified using wearable sensors,²¹ as well as the time of force exertion, which was freely determined by the subject, varying between 1 and 5 seconds.

There was a perceived subjectivity in the force applied between medium and low force, resulting in smaller differences in the measured angle. This could also be due to the resistance of the dynamometer, as shown in Figure 2a, which allows more angle variation at lower resistance, as seen in the resulting calculated angle. Finally, the type of grip was always the same with the dynamometer form, this can be evaluated in future research by determining EMG and FMG variations and more complex angle and force prediction with different grip types and weights,²² considering the complexity and degrees of freedom of the hand.²³

5. CONCLUSIONS AND FUTURE WORK

The complementarity of EMG and FMG signals was demonstrated by comparing six commonly used machine learning classifiers. In addition, precise estimation of force and angle to improve hand prostheses' control was possible using machine learning techniques, sEMG sensors, and optical fiber sensors. The results with the RF algorithm showed optimal performance for angle and force estimation, however, this work leads to future work on solving some identified limitations. First, the application of this approach requires the expansion of the dataset to include a larger and more diverse sample of non-amputee and amputee subjects, generating a higher variability of individual characteristics, such as forearm and hand size, perceived fatigue, and gender, to find out if they are determinant characteristics in classification or force performance (e.g., hand size could affect the grip on the dynamometer).

In addition, further analysis may lead to the extraction of more features of interest, providing more information to the models and leading to better results in other evaluated models. Finally, control tests with the prosthesis can be performed using the selected algorithm to determine the real-time estimation and accuracy for activating the device. Also, amputees may have a different myoelectric response, which could lead to changes in

the setup and model, for which future work could include tests with pathological subjects and evaluation of the prosthesis with patients, promoting training in control through motivational tools (i.e. serious games).

ACKNOWLEDGMENTS

This work is partially supported by FAPES (EXTRATO ATA DA 6^a REUNIÃO CCAF/2022 – “CPID 2.2”, 633/2022), CNPq (310668/2021-2 and 404111/2023-8), Finep (Tec assistiva - 2784/20 and Tec assistiva 2 - 2132/22), and IEEE RAS SPARX.

REFERENCES

- [1] Pierrie, S. N., “Current concepts in upper-extremity amputation,” *The Journal of Hand Surgery* **43**, 657–667 (Jul 2018).
- [2] Major, M. J., “Effects of upper limb loss or absence and prosthesis use on postural control of standing balance,” *American Journal of Physical Medicine and Rehabilitation* **99**, 366–371 (May 2020).
- [3] Calabrese, L., Maffoni, M., Torlaschi, V., and Pierobon, A., “What is hidden behind amputation? quantitative systematic review on psychological adjustment and quality of life in lower limb amputees for non-traumatic reasons,” *Healthcare* **11**(11) (2023).
- [4] Yuan, B., Hu, D., Gu, S., Xiao, S., and Song, F., “The global burden of traumatic amputation in 204 countries and territories,” *Frontiers in Public Health* **11**, Not available (OCT 2023).
- [5] Shahsavari, H., Matourypour, P., Ghiyasvandian, S., Ghorbani, A., Bakhshi, F., Mahmoudi, M., and Golestannejad, M., “Upper limb amputation; care needs for reintegration to life: An integrative review,” *International Journal of Orthopaedic and Trauma Nursing* **38**, 100773 (aug 2020).
- [6] Robbie Brack, E. H. A., “A review of technology, materials and ramp;d challenges of upper limb prosthesis for improved user suitability,” *Journal of Orthopaedics* **23**, 88–96 (Jan 2021).
- [7] Robbie Brack, E. H. A., “A review of technology, materials and ramp;d challenges of upper limb prosthesis for improved user suitability,” *Journal of Orthopaedics* **23**, 88–96 (Jan 2021).
- [8] Gloumakov, Y., “Trajectory control—an effective strategy for controlling multi-dof upper limb prosthetic devices,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering* **30**, 420–430 (2022).
- [9] Li, T., “An artificial intelligence-motivated skin-like optical fiber tactile sensor,” *Advanced Intelligent Systems* **5** (Aug 2023).
- [10] Gan, L., “Development of a fiber bragg grating-based force sensor for minimally invasive surgery—case study of ex-vivo tissue palpation,” *IEEE Transactions on Instrumentation and Measurement* **72**, 1–12 (2023).
- [11] Franzke, A. W., “Users’ and therapists’ perceptions of myoelectric multi-function upper limb prostheses with conventional and pattern recognition control,” *PLOS ONE* **14**, e0220899 (Aug 2019).
- [12] Chen, Z., “A multimodal multistream multilevel fusion network for finger joint angle estimation with hybrid semg and fmg sensing,” *Alexandria Engineering Journal* **110**, 9–23 (Jan 2025).
- [13] Roberto Díaz-Amador, M. A. M.-R., “Towards the reduction of the effects of muscle fatigue on myoelectric control of upper limb prostheses,” *DYNA* **86**, 110–116 (Jan 2019).
- [14] Choi, Y., Lee, S., Sung, M., Park, J., Kim, S., and Choi, Y., “Development of emg-fmg based prosthesis with pvdf-film vibrational feedback control,” *IEEE Sensors Journal* **21**, 23597–23607 (OCT 2021).
- [15] Briko, A., Kapravchuk, V., Kobelev, A., Hammoud, A., Leonhardt, S., Ngo, C., Gulyaev, Y., and Shchukin, S., “A way of bionic control based on ei, emg, and fmg signals,” *Sensors* **22**, 152 (DEC 2021).
- [16] Jason S. Gharibo, M. D. N., “Multi-modal prosthesis control using semg, fmg and imu sensors,” *2022 44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2983–2987 (Jul 2022).
- [17] Cortes, F. R., “Development of a force myography sensor for prhand prosthesis activation using fiber bragg grating sensor and 3d printing,” *2024 Latin American Workshop on Optical Fiber Sensors (LAWOFS)*, 1–2 (May 2024).
- [18] Miften, F. S., “A new framework for classification of multi-category hand grasps using emg signals,” *Artificial Intelligence in Medicine* **112**, 102005 (Feb 2021).

- [19] Jiang, X., “Force exertion affects grasp classification using force myography,” *IEEE Transactions on Human-Machine Systems* **48**, 219–226 (Apr 2018).
- [20] Arco, L. D., “The prhand: Functional assessment of an underactuated soft-robotic prosthetic hand,” *2022 9th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)* , 1–6 (Aug 2022).
- [21] Otálora, S., “Data-driven approach for upper limb fatigue estimation based on wearable sensors,” *Sensors* **23**, 9291 (Nov 2023).
- [22] Xu, B., “Natural grasping movement recognition and force estimation using electromyography,” *Frontiers in Neuroscience* **16** (Oct 2022).
- [23] Zhaojie Ju, H. L., “Human hand motion analysis with multisensory information,” *IEEE/ASME Transactions on Mechatronics* **19**, 456–466 (Apr 2014).